
2: Basics of Probability

SC6116: Game Theory and Blockchain

Prof Zhang Jie

Topics Covered

- Some useful concepts of probability calculations
 - Confusion matrix
 - Bayesian statistics
 - The Brier score
- Estimate the probability of occurrence of some events (e.g. in decision theory)
- Assess the quality of a prediction (e.g. of a blockchain-based prediction market).

Confusion Matrix

■ Quality of prediction

	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	a	b
	Class=No	c	d

a: TP (true positive)
b: FN (false negative)
c: FP (false positive)
d: TN (true negative)

Confusion Matrix

Actual	Prediction
Yes	Yes
No	No
No	Yes
Yes	No
Yes	Yes
Yes	Yes
No	No
No	Yes
Yes	No
Yes	Yes
...	...

a++
d++
c++
b++

	PREDICTED CLASS		
		Class=Yes	Class=No
	Class=Yes	a	b
	Class=No	c	d

a: TP (true positive)

b: FN (false negative)

c: FP (false positive)

d: TN (true negative)

Example

ACTUAL CLASS	PREDICTED CLASS	
		Class=spam Class=Normal
	Class=spam	40 (a) 60 (b)
	Class=Normal	160 (c) 5000 (d)

- What is the likelihood that an email predicted to be spam actually is a spam?
- Precision or Positive Prediction Value (PPV)

$$PPV = \frac{tp}{tp + fp}$$

- Negative Prediction Value (NPV)

$$NPV = \frac{tn}{tn + fn}$$

Example

- Sensitivity (SE) or true positive rate (TPR)

$$SE = \frac{tp}{tp + fn}$$

- Specificity (SP) or true negative rate (TNR)

$$SP = \frac{tn}{tn + fp}$$

- Positive and negative likelihood ratio

$$LR_+ = \frac{Sensitivity}{1 - Specificity} \quad LR_- = \frac{1 - Sensitivity}{Specificity}$$

- The positive likelihood ratio tells us how the likelihood of a positive result changes if the test result is positive; the negative likelihood ratio tells us how the likelihood of a positive result changes if the test result is negative

Bayes/Conditional Probability

- A probabilistic framework for classification problems
- Conditional Probability:

$$P(C | A) = \frac{P(A, C)}{P(A)} \quad (1)$$

$$P(A | C) = \frac{P(A, C)}{P(C)} \quad (2)$$


- **Bayes Theorem:** replace $P(A, C)$ in (1) with $P(A|C)*P(C)$ from (2) – the only difference between (1) and (3)

$$P(C | A) = \frac{P(A | C)P(C)}{P(A)} \quad (3)$$

Example

- Given: $P(C | A) = \frac{P(A | C)P(C)}{P(A)}$
- A doctor knows that meningitis* causes stiff neck 50% of the time
 - Prior probability of any patient having meningitis is 1/50,000
 - Prior probability of any patient having stiff neck is 1/20
- If a patient has stiff neck, what's the probability he/she has meningitis?

Given class C=meningitis, A=stiff neck, compute $P(C|A)$

$$P(C | A) = \frac{P(A | C)P(C)}{P(A)} = \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

Bayes Theorem is an interesting formula: we can use $P(A|C)$ (and other factors) to compute $P(C|A)$, which may be hard to directly estimate

Brier Score

- Proper score function measures the accuracy of a probability prediction

- f_t : probability that was forecast
- o_t : actual outcome of the event at instance t
- N : Number of forecasting instances
- t : instance

$$BS = \frac{\sum_{t=1}^N (f_t - o_t)^2}{N}$$

- Multi-category forecasts

- R : number of possible classes in which the event can fall. $R = 2$ for the case rain / no rain, $R = 3$ for the case long, normal, short.
- N : number of overall instances of all classes
- f_{ti} : predicted probability of class i
- o_{ti} : observation for instance t and class i . $o_{ti} = 1$ if this is the i th class, otherwise it is 0.

$$BS = \frac{\sum_{t=1}^N \sum_{i=1}^R (f_{ti} - o_{ti})^2}{N}$$

Brier Skill Score (BSS)

- Compare with the score of a reference method

$$BSS = 1 - \frac{BS}{BS_{ref}}$$

- BS_{ref} : Brier score for a reference forecasting method

Index	1	2	3	4	5
Event	0	0	1	1	1
Forecasted Probability	0.1	0.2	0.6	0.8	0.9
Square Error with f	0.01	0.04	0.16	0.04	0.01

- $BS = 0.052$
- The reference model assumes a probability of 0.30 throughout every day.
 - $BS_{ref} = 0.33$
- $BSS = 0.84242$